Vignesh Subramanian

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Research Interests

I am a Ph.D. student at Georgia Institute of Technology working on reliable generalization in reinforcement learning. My research combines formal methods, neuro-symbolic reasoning, and LLMs to build verifiable agents that can justify and certify their decisions in safety-critical settings.

Education

Georgia Institute of Technology **Aug 2023 – Ongoing**
Ph.D. in Computer Science | 4th Year | GPA: 3.9 Georgia, USA
Advisor – Prof. Suguman Bansal | *Anticipated Graduation: May 2027*

National Institute of Technology, Tiruchirappalli **Aug 2019 - April 2023**
B.Tech. - Electrical and Electronics Engineering Tiruchirappalli, Tamil Nadu
Minor - Computer Science Engineering

Relevant Coursework: Logic in Computer Science, SAT/SMT solvers, Machine Learning and Deep Learning, Software Analysis and Testing, Artificial Intelligence, Data Structures and Algorithms

Publications

Decoupled Behavioral Cloning for Scalable Inductive Generalization in RL from Specifications
*Neural Information Processing Systems, 2026 (**Under Submission**)*
Vignesh Subramanian, Subhajit Roy, Suguman Bansal

Certificate-Guided Evaluation of Reinforcement Learning Generalization
*Neural Information Processing Systems, Evaluation and Datasets Track, 2026 (**Under Submission**)*
Vignesh Subramanian, Djordje Zikelic, Suguman Bansal

Inductive Generalization in Reinforcement Learning from Specifications
Automated Technology for Verification and Analysis, 2025
Vignesh Subramanian, Rohit Kushwah, Subhajit Roy, Suguman Bansal **[Link]**

Reinforcement Learning for Stochastic Max-Plus Linear Systems
Conference on Decision and Control, IEEE 2023
Vignesh Subramanian, Farzaneh Farhadi, Sadegh Soudjani **[Link]**

A Novel Facial Emotion Recognition Model Using Segmentation VGG-19 Architecture
International Journal of Information Technology, Springer, 2023
Vignesh Subramanian, Savithadevi, M. Sridevi, Rajeswari Sridhar **[Link]**

SIHeDA-Net: Sensor to Image Heterogeneous Domain Adaptation for Sign Language Detection
Medical Imaging with Deep Learning, 2022
Ishikaa Lunawat, **Vignesh Subramanian**, S. P. Sharan **[Link]**

Teaching Experience

CS 3600 - Artificial Intelligence **Spring 2026**
Georgia Institute of Technology *Georgia, USA*

CS 8803 - Formal Methods in Reinforcement Learning **Fall 2025**
Georgia Institute of Technology *Georgia, USA*

CS 8803 - SAT/SMT Solvers **Fall 2024**
Georgia Institute of Technology *Georgia, USA*

Internships

Newcastle University, United Kingdom | Paper link **Jan 2022 - Nov 2022**
Research Intern - Mentor: Prof. Sadegh Soudjani *Newcastle, England*

- Proposed a novel optimization strategy using Deep Q-Learning for Stochastic Max-Plus-Linear Discrete Event Systems under uncertainties, achieving a 2.5x speedup over Model Predictive Control for minimizing stochastic delays in railway systems, published at **Conference on Decision and Control, IEEE, 2023**.

Samsung R&D Institute, Bangalore | Report link **Aug 2021 - Feb 2022**
Research Intern - Mentor: Mr. Sujoy Saha, Mr. Rajat Kumar Jain *Karnataka, India*

- Developed a Tri-Stage Occlusion Handling Normal Map Estimation Algorithm to enhance 3D human normal map estimation on blurred, noisy, low-res 2D images, achieving 92.78% IoU (vs. 71.26% baseline) by effectively addressing occlusions from shadows, blur, and background.

National Institute of Technology, Tiruchirappalli | Paper link | GitHub link **Feb 2021 - Dec 2021**
Research Intern - Mentor: Prof. M. Sridevi *Tamil Nadu, India*

- Designed a novel CNN architecture integrating U-Net and VGG layers for Facial Emotion Recognition, achieving 75.97% accuracy on the FER-2013 dataset, ranking in the **Top 5** on the global leaderboard. Findings were published in the **International Journal of Information Technology, Springer**.

Technical Skills

- **Languages**: C/C++, Python, Java, JavaScript, Lean
- **Packages/Frameworks**: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, OpenAI Gym, Stable Baselines
- **Reinforcement Learning**: Policy Gradient Methods, Generalization Algorithms, Safety Certificates for RL, Reachability and Safety Analysis
- **Formal Verification**: Model Checking, Formal Methods for Safety Verification
- **LLM**: Pre-training and fine-tuning large-scale transformer models, LLM-based inference for complex logical queries, reinforcement learning from human feedback (RLHF), and automated proof generation.

Awards and Talks

Research Talk **2026**
PLSEFM + AI Workshop, Georgia Institute of Technology *Georgia, USA*

Research Talk **2026**
Indian Institute of Technology Kanpur *Kanpur, India*

Graduate Poster Symposium – Third Place **2024**
Junior Student Category, Georgia Institute of Technology *Georgia, USA*

NeurIPS Travel Grant Award **2023**
Neural Information Processing Systems *Louisiana, USA*

Positions Of Responsibility

Reviewer **2025**
AAAI Workshop on Generalization in Planning.

Student Research Mentor, Machine Learning and AI **2020 – 2023**
Spider R&D, NIT Trichy — Led ML/AI projects; built SIHeDA-NET (Paper) and GISiL (Demo).

Coordinator, Workshops and Publicity **2020 – 2021**
Currents Symposium, NIT Trichy — Managed workshops, publicity; hosted 600+ participants.